Technological Requirements
&
Learning Plan

COMP3095 – Capstone Project 1

QuietSpace

Group 3

Team Members:

Baris Isci – 101531238

Daniil Orlov - 101500729

Ibrahim Yondem - 101525165

Gozde Baran - 101515982

Nguyen Minh Triet Luu – 101542519

Instructor:

Anjana Shah

PART I -- MockUp/Prototype

Developer Mode link:
https://www.figma.com/design/xWWkaweOQCgGM1NY5Y1RBp/Capstone_Prototype_main_branch?node-id=0-1&m=dev&t=1RCFzQSXyAbq2peY-1

PART II -- Technology Requirements

A) Technological Requirements

Aspect	Selected Tools	Pros	Cons
Languages	*Kotlin*	Kotlin is a modern, concise, and fully supported Android language that integrates seamlessly with Jetpack Compose, reducing boilerplate and speeding up UI development. It's officially endorsed by Google and has strong community support, making it ideal for Android-first projects; also, the team has prior experience working with it	Kotlin development is limited to the Android ecosystem and requires learning Jetpack Compose and advanced concepts such as Coroutines and dependency injection to fully utilize its potential
	React Native (potential next stage)	React Native enables cross-platform development using a single codebase for Android, iOS, and web, reducing future maintenance and ensuring faster multi-platform deployment. Its modular structure and large open-source community make it a good long-term scalability option	The team currently lacks experience with React Native, which would require learning JavaScript and the React ecosystem. Rebuilding existing Kotlin code and managing native modules would increase development effort and risk inconsistencies across platforms
Database	*Option 1, SQL: Supabase / private server*	A SQL database aligns with the project's normalized relational model and works directly with Android Room, supporting complex queries and relationships. Supabase provides a managed PostgreSQL solution that simplifies hosting, integrates authentication, and ensures strong data integrity	Setting up and maintaining an SQL backend requires knowledge of database administration, migrations, and hosting. For small teams, configuration, schema changes, and maintaining a live server may add unnecessary overhead
	Option , NoSQL2: Firestore/MongoDB	Firestore offers an easy-to-use cloud-hosted NoSQL solution that integrates tightly with the currently used Firebase Authentication and Storage. It automatically handles scaling, provides real-time updates, and eliminates the need to manage server infrastructure. The free packages offer enough functionality for our initial purposes, which eliminates unnecessary expenses	NoSQL databases require denormalization, which can lead to data duplication and inconsistent updates, as well as create extra workload matching with the local storage (Android Room). Complex queries and aggregate operations (like averages and counts) are harder to implement and may require Cloud Functions or extra logic in the app layer

Hardware	*Option 1: Third-party services*	Using cloud services simplifies deployment and eliminates hardware management. These platforms provide high uptime, automatic scaling, and built-in security, freeing the team to focus entirely on app development	Dependence on external providers introduces ongoing costs beyond free tiers and limits customization. The team also has less control over backend configurations, data ownership, and long-term flexibility
	Option 2: Hosting a private server	Running the server locally gives full control over configurations, storage, and access permissions. It's a great learning opportunity to understand backend infrastructure and can be demonstrated easily during testing or presentations, and does not require any upfront expenses	Self-hosting requires technical setup (databases, SSL, firewalls) and continuous maintenance. It's less reliable due to hardware and network limitations, and downtime could disrupt the project during development or evaluation. Also, a much less feasible long-term solution
Version Control	*Git, GitHub*	Widely used version control system supporting branching, merging, and collaborative workflows. GitHub provides remote repositories, issue tracking, and pull requests for seamless teamwork and code review	Requires understanding of Git commands and workflows; merge conflicts can occur if branching and code review isn't done properly
Team Coordination	*Jira*	Powerful project management tool supporting agile workflows, sprint planning, and task tracking. Helps the team organize work, assign responsibilities, and monitor progress effectively through dashboards and issue boards	Requires initial setup and acclimatization; may feel complex or heavy for the team if not configured properly

B) Learning Plan

Daniil Orlov – Team Leader

Skill/Technology	Current Knowledge (%)	Expected Knowledge (%)	Comment/Short Explanation	Start Date	Target Date	Resources
PostgreSQL (Relational DB)	*70%*	85%	Gain basic SQL knowledge to ensure data consistency in tests	*Nov-25*	Mar-26	Udemy, postgresql.org, YouTube
MongoDB (NoSQL DB)	*70%*	85%	Understand NoSQL paradigm (comparable to Firebase Firestore) for flexible data modeling	*Nov-25*	Mar-26	Udemy, GBC COMP 3095, mongodb.com, YouTube
React Native (Mobile UI)	*50%*	75%	Explore cross-platform UI development for potential iOS/web expansion	*Nov-25*	Mar-26	Udemy, reactnative.dev, YouTube
Java / Kotlin	*70%*	90%	Understand app's UI to design effective front-end	*Nov-25*	Mar-26	GBC COMP3074, YouTube
Node.js & Express (Backend)	*70%*	85%	Strengthen backend development skills (Node/Express) for server-side logic	*Nov-25*	Mar-26	GBC COMP3123, YouTube
Firebase (Backend Services)	*50%*	80%	Learn Firebase basics (Firestore, Auth) to verify data and triggers	*Nov-25*	Mar-26	Udemy, firebase.google.com
DevOps Tools (Render, Netlify)	*80%*	95%	Set up CI/CD pipelines and understand deployment (Render/Netlify) for smooth releases	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Git & GitHub (Version Control)	*75%*	90%	Manage repository and enforce team Git workflow (branching, PR reviews)	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Testing & QA Tools	*65%*	85%	Oversee testing frameworks and code	*Nov-25*	Mar-26	Udemy, YouTube

Gozde Baran – Quality Assurance

Skill/Technology	Current Knowledge (%)	Expected Knowledge (%)	Comment/Short Explanation	Start Date	Target Date	Resources
PostgreSQL (Relational DB)	*70%*	85%	Reinforce SQL skills for potential Postgres (Supabase) use	*Nov-25*	Mar-26	Udemy, postgresql.org, YouTube
MongoDB (NoSQL DB)	*50%*	75%	Learn NoSQL concepts (parallel to Firestore's data model) for unstructured data testing	*Nov-25*	Mar-26	Udemy, GBC COMP 3095, mongodb.com, YouTube
React Native (Mobile UI)	*60%*	85%	Understand app's UI to design effective front-end test cases	*Nov-25*	Mar-26	Udemy, reactnative.dev, YouTube
Java / Kotlin	*70%*	85%	Understand app's UI to design effective front-end test cases	*Nov-25*	Mar-26	GBC COMP3074, YouTube
Node.js & Express (Backend)	*70%*	85%	Familiarize with server logic to create relevant API test scenarios	*Nov-25*	Mar-26	GBC COMP3123, YouTube
Firebase (Backend Services)	*50%*	70%	Learn Firebase basics (Firestore, Auth) to verify data and triggers	*Nov-25*	Mar-26	Udemy, firebase.google.com
DevOps Tools (Render, Netlify)	*80%*	95%	Understand deployment process to support end-to-end testing on the deployed app	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Git & GitHub (Version Control)	*75%*	95%	Use version control for test scripts and collaborate on code changes	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Testing & QA Tools	*75%*	95%	Master test planning, bug tracking, and	*Nov-25*	Mar-26	Udemy, YouTube

Baris Isci – UI Developer (Front-End Specialist)

Skill/Technology	Current Knowledge (%)	Expected Knowledge (%)	Comment/Short Explanation	Start Date	Target Date	Resources
PostgreSQL (Relational DB)	*50%*	75%	Gain basic SQL knowledge to ensure data consistency in tests	*Nov-25*	Mar-26	Udemy, postgresql.org, YouTube
MongoDB (NoSQL DB)	*50%*	75%	Learn NoSQL concepts (parallel to Firestore's data model) for unstructured data testing	*Nov-25*	Mar-26	Udemy, GBC COMP 3095, mongodb.com, YouTube
React Native (Mobile UI)	*75%*	90%	Develop cross-platform UI skills with React Native for broader front-end expertise	*Nov-25*	Mar-26	Udemy, reactnative.dev, YouTube
Java / Kotlin	*75%*	95%	Understand app's UI to design effective front-end test cases	*Nov-25*	Mar-26	GBC COMP3074, YouTube
Node.js & Express (Backend)	*50%*	85%	Learn fundamental Node.js/Express concepts to integrate frontend with backend APIs	*Nov-25*	Mar-26	GBC COMP3123, YouTube
Firebase (Backend Services)	*50%*	75%	Implement Firebase integration (Firestore queries, Storage) in the app	*Nov-25*	Mar-26	Udemy, firebase.google.com
DevOps Tools (Render, Netlify)	*60%*	85%	Understand deployment process to support end-to-end testing on the deployed app	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Git & GitHub (Version Control)	*75%*	95%	Use GitHub for code versioning and collaborate with the team on feature branches	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Testing & QA Tools	*55%*	80%	Apply unit and UI testing (JUnit, Espresso) to ensure UI features work correctly	*Nov-25*	Mar-26	Udemy, YouTube

Ibrahim Yondem – UI Developer (Front-End Specialist)

Skill/Technology	Current Knowledge (%)	Expected Knowledge (%)	Comment/Short Explanation	Start Date	Target Date	Resources
PostgreSQL (Relational DB)	*50%*	75%	Gain basic SQL knowledge to ensure data consistency in tests	*Nov-25*	Mar-26	Udemy, postgresql.org, YouTube
MongoDB (NoSQL DB)	*50%*	75%	Learn NoSQL concepts (parallel to Firestore's data model) for unstructured data testing	*Nov-25*	Mar-26	Udemy, GBC COMP 3095, mongodb.com, YouTube
React Native (Mobile UI)	*75%*	95%	Develop cross-platform UI skills with React Native for broader front-end expertise	*Nov-25*	Mar-26	Udemy, reactnative.dev, YouTube
Java / Kotlin	*75%*	95%	Understand app's UI to design effective front-end test cases	*Nov-25*	Mar-26	GBC COMP3074, YouTube
Node.js & Express (Backend)	*50%*	85%	Utilize Firebase services (Firestore, Auth) effectively in front-end development	*Nov-25*	Mar-26	GBC COMP3123, YouTube
Firebase (Backend Services)	*50%*	75%	Implement Firebase integration (Firestore queries, Storage) in the app	*Nov-25*	Mar-26	Udemy, firebase.google.com
DevOps Tools (Render, Netlify)	*60%*	85%	Understand deployment process to support end-to-end testing on the deployed app	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Git & GitHub (Version Control)	*75%*	95%	Use GitHub for code versioning and collaborate with the team on feature branches	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Testing & QA Tools	*50%*	80%	Apply unit and UI testing (JUnit, Espresso) to ensure UI features work correctly	*Nov-25*	Mar-26	Udemy, YouTube

Nguyen Minh Triet Luu (Felix) – Back-End Developer

Skill/Technology	Current Knowledge (%)	Expected Knowledge (%)	Comment/Short Explanation	Start Date	Target Date	Resources
PostgreSQL (Relational DB)	*70%*	90%	Design and manage relational databases (Postgres) for alternative backend (Supabase)	*Nov-25*	Mar-26	Udemy, postgresql.org, YouTube
MongoDB (NoSQL DB)	*75%*	95%	Apply NoSQL concepts (akin to Firestore) for flexible data management when needed	*Nov-25*	Mar-26	Udemy, GBC COMP 3095, mongodb.com, YouTube
React Native (Mobile UI)	*55%*	80%	Develop cross-platform UI skills with React Native for broader front-end expertise	*Nov-25*	Mar-26	Udemy, reactnative.dev, YouTube
Java / Kotlin	*60%*	85%	Understand app's UI to design effective front-end test cases	*Nov-25*	Mar-26	GBC COMP3074, YouTube
Node.js & Express (Backend)	*80%*	98%	Develop robust backend logic using Node.js/Express (incl. Firebase Cloud Functions	*Nov-25*	Mar-26	GBC COMP3123, YouTube
Firebase (Backend Services)	*75%*	95%	Master Firebase backend (Firestore, Auth, Cloud Functions) for core app functionality	*Nov-25*	Mar-26	Udemy, firebase.google.com
DevOps Tools (Render, Netlify)	*70%*	95%	Implement deployment processes/CI for backend code (e.g. testing Cloud Functions, explore Render)	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Git & GitHub (Version Control)	*75%*	95%	Use GitHub for code versioning and collaborate with the team on feature branches	*Nov-25*	Mar-26	GBC COMP3104, YouTube
Testing & QA Tools	*60%*	85%	Perform thorough backend testing (Cloud Function tests, integration tests with the app)	*Nov-25*	Mar-26	Udemy, YouTube